Amiratabak Bahengam

Ph.D. Student | Graduate Research Assistant, Department of Systems Engineering
Stevens Institute of Technology, Hoboken, NJ, USA
abahenga@stevens.edu | +1 929 326 0266 | bahengam.me
ORCID: 0009-0009-5587-5000 | Google Scholar

Research Summary

Ph.D. student in Systems Engineering developing hybrid quantum–classical optimization methods for large-scale complex engineering problems. Contributions span Hamiltonian formulations for constrained design optimization, decomposition-based QAOA for problems beyond the reach of near-term quantum hardware, and mixed-integer programming for space vehicle routing. Published in *IEEE Access* and presented at INFORMS, AIAA, ASME IDETC-CIE, USACM, and the Mixed Integer Programming (MIP) Workshop.

Education

Columbia University — **New York, NY, USA**
Graduate Coursework (non-degree) — Quantum Computing and Quantum Machine Learning — Sept. 2025 – May 2026

Stevens Institute of Technology — **Hoboken, NJ, USA**
Ph.D. in Systems Engineering — GPA: 4.00/4.00 — Sept. 2023 – Dec. 2029 (expected)
Dissertation focus: Quantum computing applications to complex engineering design problems.
Advisor: Dr. Hao Chen
M.Eng. in Systems Engineering — GPA: 3.97/4.00 — Sept. 2023 – May 2025
Advisor: Dr. Hao Chen

University of Tehran — **Tehran, Iran**
B.Sc. in Industrial Engineering — GPA: 3.81/4.00 — Sept. 2018 – June 2023
Advisor: Dr. Seyed Farid Ghaderi
Coursework highlights: Operations Research, Systems Analysis, Engineering Statistics, Fundamentals of Probability, Linear Algebra, Decision Analysis

Appointments

Stevens Institute of Technology — Department of Systems Engineering — **Hoboken, NJ, USA**
Graduate Research Assistant — Sept. 2023 – Present
Quantum optimization for aerospace, space logistics, and complex engineering systems design. Principal Investigator: Dr. Hao Chen.

Research Interests

- Quantum computing for large-scale systems optimization
- Hybrid quantum–classical optimization algorithms for engineering applications
- Decomposition methods for mixed-integer programming on near-term quantum hardware
- Quantum algorithm development for engineering design and decision-making
- Quantum circuit optimization and noise mitigation strategies
- Quantum machine learning for data-driven decision-making and system modeling
- Quantum-assisted mission planning and optimization in space logistics

Peer-Reviewed Journal Articles

1. D. Sisodiya, **A. Bahengam**, H. W. Lee, and H. Chen, "Decomposition-Based QAOA for Maximum Coverage Location Problem in Satellite Constellation Design," *IEEE Access*, early access, Oct. 2026. Open access. doi:10.1109/ACCESS.2026.3739711
2. A. Tajally, M. Z. Vamarzani, M. Ghanavati-Nejad, F. R. Zeynali, M. Abbasian, and **A. Bahengam**, "A hybrid machine learning-based decision-making model for viable supplier selection problem considering circular

economy dimensions," *Environment, Development and Sustainability*, Jan. 2025. doi:10.1007/s10668-025-06014-9

Peer-Reviewed Conference Proceedings

1. **A. Bahengam**, M.-A. Miri, R. J. Rupert, W. Dyk, and H. Chen, "Quantum-Assisted Space Logistics Mission Planning," *AIAA SciTech 2025 Forum*, Orlando, FL, USA, no. AIAA-2025-1107, Jan. 2025. doi:10.2514/6.2025-1107

Manuscripts Under Review

1. **A. Bahengam** and H. Chen, "A Hamiltonian-Based Quantum Formulation for Multi-objective Engineering Systems Design Optimization," *under peer review.*

Conference Presentations and Extended Abstracts

1. **A. Bahengam** and H. Chen, "Two-Stage Entropy Quantum Optimization of Propellant-Aware Space Logistics," *2026 INFORMS Annual Meeting*, San Francisco, CA, USA, Nov. 2026.
2. **A. Bahengam** and H. Chen, "Entropy Quantum Optimization of Time-Expanded Space Logistics Mission Planning," *USACM Workshop on Quantum Computing for Engineering Applications (QC2026)*, Nashville, TN, USA, Aug. 2026.
3. R. Talebi Khameneh, **A. Bahengam**, and H. Chen, "A Propellant-Coupled Space Vehicle Routing MILP and Its Hybrid Quantum–Classical Decomposition," *2026 Mixed Integer Programming (MIP) Workshop*, Stamford, CT, USA, May 2026.
4. **A. Bahengam** and H. Chen, "Analyzing Technology Evolvement and Innovation Pattern in Engineering Systems Development," extended abstract, *ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2024)*, IDETC2024-148323, Washington, DC, USA, Aug. 2024.
5. **A. Bahengam** and H. Chen, "Hamiltonian Representation for Engineering Systems Design," extended abstract, *ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2024)*, IDETC2024-148307, Washington, DC, USA, Aug. 2024.

Research Experience

- **Decomposition-Based QAOA for Satellite Constellation Design.** Developed a decomposition strategy that partitions the maximum coverage location problem into subproblems tractable on near-term quantum processors, enabling constellation design instances beyond the reach of direct QAOA encoding. *(Published, IEEE Access, 2026)*
- **Propellant-Coupled Space Vehicle Routing.** Formulated a mixed-integer linear program coupling routing decisions to propellant consumption, and designed a hybrid quantum–classical decomposition to solve it. *(Presented at the 2026 MIP Workshop)*
- **Hamiltonian Representation of Engineering Systems Design.** Proposed a Hamiltonian-based quantum optimization method for complex design problems; designed QAOA-inspired circuits for multi-objective constrained optimization. *(Manuscript under peer review)*
- **Quantum-Assisted Space Logistics Mission Planning.** Modeled space logistics mission planning as a quantum-solvable optimization problem and benchmarked hybrid solution approaches. *(Published, AIAA SciTech 2025)*
- **Hybrid Machine Learning for Sustainable Supplier Selection.** Integrated machine learning with circular economy metrics for supplier evaluation; achieved >93% accuracy on a real-world automotive case study. *(Published, Environment, Development and Sustainability)*
- **Technology Evolvement Analysis in Aerospace Systems.** Applied natural language processing to 356,000+ NASA technical reports using Zipf's and Heaps' laws, revealing long-term patterns of innovation and knowledge growth in aerospace research. *(Extended abstract, IDETC-CIE 2024)*
- **Systems Thinking for Quantum Computing Evolution.** Applied systemigrams, causal loop diagrams, and stakeholder mapping to assess systemic drivers and impacts of quantum technology adoption. *(In progress)*
- **B.Sc. Thesis — Sustainable Supplier Selection under Industry 4.0.** Developed a decision framework

integrating circular economy principles to advance sustainability and digitalization in supply chain operations.

Professional Service and Peer Review

- **Technical Reviewer**, *AIAA ASCEND 2026*, Washington, DC, USA (19–21 May 2026). Invited by the technical program committee; completed peer review of **37 abstract submissions** across space systems, propulsion, and in-situ resource utilization tracks, scoring each on technical quality, originality, relevance, and clarity, and issuing accept/reject recommendations. *Oct. 2025*

Awards, Honors, and Scholarships

- **Best Translation of a Quantum Concept**, Quantum Design Hackathon, Unitary Foundation (2026). Awarded for the clearest translation of a technical quantum computing concept into an accessible design artifact, judged against submissions from an international field of participants.
- **Best Scientific Process and Interpretation**, *A Real Quantum Hackathon*, Harmoniqs (2026). Awarded for rigor of experimental methodology and quality of interpretation of quantum computational results.
- **Provost Doctoral Fellowship**, Stevens Institute of Technology, USA (2023–present). Four-year full fellowship awarded for academic excellence in the Systems Engineering Ph.D. program.
- **Dean's List**, University of Tehran, Iran (2019–2023). Placed among the top 25% of students by GPA in the graduating class.
- **National University Entrance Examination Distinction**, National Organization for Educational Testing, Iran (Aug. 2018). Ranked in the top 4% of 144,437 participants nationally.
- **Undergraduate Merit Scholarship**, University of Tehran, Iran (2018–2023). Full tuition exemption for four years of B.Sc. study, awarded on academic merit.

Certifications and Professional Training

- **Associate Systems Engineering Professional (ASEP)**, International Council on Systems Engineering (INCOSE). Awarded on passing the INCOSE knowledge examination based on the *INCOSE Systems Engineering Handbook*.
- **Quantum Excellence**, Qiskit Global Summer School 2026, IBM (Aug. 2026). Highest completion tier of IBM's quantum computing summer school, awarded on performance across the program's lecture, laboratory, and assessment track. Verifiable at credly.com/go/ormwzjhM.

Professional Memberships

- **American Institute of Aeronautics and Astronautics (AIAA)** — Student Member (current).
- **Institute for Operations Research and the Management Sciences (INFORMS)** — Student Member (current).

Technical Skills

- **Quantum Algorithms:** QAOA, VQE, Grover's algorithm, amplitude amplification.
- **Quantum Machine Learning:** variational quantum classifiers, quantum kernel methods, parameterized quantum circuits, quantum feature maps, hybrid QML pipelines.
- **Quantum Modeling & Optimization:** Hamiltonian formulation, Ising and QUBO models, quantum annealing, hybrid quantum–classical decomposition methods.
- **Noise & Error Handling:** quantum error correction codes, circuit optimization, noise mitigation.
- **Quantum Frameworks & Platforms:** Qiskit, Cirq, PennyLane, D-Wave Ocean SDK; IBM Quantum, Google Quantum AI.
- **Computational Methods:** linear and nonlinear optimization, mixed-integer programming, mathematical modeling, numerical algorithms, stochastic and discrete-event simulation, game theory, graph theory, network science.
- **Programming:** Python, R, C++, MS SQL Server, HTML/CSS.
- **Software:** MATLAB, GAMS, CPLEX, Gurobi, MS Project, MS Excel, Stata, SPSS, Tableau, Adobe Illustrator, Adobe Photoshop.